

1) Problem 2.1

3.3) given: $\vec{r} = r\hat{e}_r = \vec{r}$
 $(\frac{d}{dt}\vec{r})^2 = (\frac{d}{dt}r\hat{e}_r)^2 = v^2$

show: $v^2 = \dot{r}^2 + r^2\dot{\theta}^2$

$$(\vec{r} \cdot \vec{v})^2 = (\vec{r} \cdot \frac{d\vec{r}}{dt})^2 = r^2 v^2$$

$$(\vec{r} \cdot \vec{v})^2 = (\vec{r} \cdot \dot{\vec{r}})^2 = r^2 \dot{\theta}^2$$

$$v^2 = \dot{r}^2 + r^2\dot{\theta}^2$$

In physical terms, $\dot{r}$ = radial velocity
 $r\dot{\theta}$ = tangential velocity

$$T = 0 + h \quad \frac{1}{2}v^2 = f(r) + h$$

$$\frac{1}{2}(r\dot{\theta}^2 + \dot{r}^2) = f(r) + h$$

$$\frac{1}{2}(r\dot{\theta}^2 + \dot{r}^2) = r^2 [f(r) + h]$$

$$r\dot{\theta}^2 + \dot{r}^2 = 2r^2 [f(r) + h]$$

4) show: if $0 < e < 1$ or $e > 1$

$$|a| |k^2 - 1| = 0$$

$$a = \frac{r_p + r_a}{2}$$

$$r = \frac{a(1-e^2)}{1+e\cos f} = e(\frac{a}{r_e} - r\cos f)$$

$$r_p = \frac{a(1-e)}{1-e} \quad r_a = \frac{a(1+e)}{1+e}$$

$$a = \frac{1}{2}(\frac{r_p}{1-e} + \frac{r_a}{1+e})$$

$$2a(1-e) = -\frac{a(1+e)}{(1-e)(1+e)}$$

$$2a(1-e) = -\frac{a}{1-e^2}$$

$$2a(1-e^2) = -a$$

5.2) given: $A_{eff} = \pi a^2 (1-e^2)^{1/2} \quad \dot{A} = c/2$ if $c \neq 0$

show: if $0 < e < 1$ or $e > 1$, $P = (\frac{c}{2\pi}) a^{3/2}$

$$A = c/2P$$

$$P = \frac{2A}{c} = \frac{2\pi a^2 (1-e^2)^{1/2}}{c} \quad \frac{c}{2\pi a^2 (1-e^2)^{1/2}} = \frac{c}{2\pi a^2 (1-e^2)^{1/2}}$$

$$= \frac{2\pi a^2 (1-e^2)^{1/2}}{c}$$

$$= \frac{2\pi a^{3/2}}{c}$$

$$P = \frac{2\pi a^{3/2}}{c}$$

5.4) define moment of inertia

$$2I = m r^2$$

$$\text{prove } \ddot{I} = 2T - U + T + h = U + 2h$$

$$I = \frac{1}{2} m r^2 = \frac{1}{2} m \dot{\vec{r}} \cdot \dot{\vec{r}}$$

$$\dot{I} = \frac{1}{2} m (\dot{\vec{r}} \cdot \dot{\vec{r}} + \vec{r} \cdot \ddot{\vec{r}})$$

$$= m \dot{\vec{r}} \cdot \dot{\vec{r}} + \frac{1}{2} m \vec{r} \cdot \ddot{\vec{r}}$$

$$\ddot{I} = m (\ddot{\vec{r}} \cdot \dot{\vec{r}} + \dot{\vec{r}} \cdot \ddot{\vec{r}}) = m (\dot{\vec{r}} \cdot \ddot{\vec{r}} + \ddot{\vec{r}} \cdot \dot{\vec{r}}) = 2m \dot{\vec{r}} \cdot \ddot{\vec{r}}$$

$$= m \dot{\vec{r}} \cdot \ddot{\vec{r}} + \frac{1}{2} m \vec{r} \cdot \ddot{\vec{r}}$$

$$= \frac{m \dot{\vec{r}} \cdot \ddot{\vec{r}}}{2} + m \left(-\frac{U}{r} \vec{r} \cdot \ddot{\vec{r}} \right)$$

$$= \frac{m \dot{\vec{r}} \cdot \ddot{\vec{r}}}{2} + m \left(-\frac{U}{r} r^2 \right)$$

$$\ddot{I} = 2T - U = 2T - (T + h)$$

$$= 2U + 2h - U = U + 2h$$

$$\ddot{I} = U + 2h = T + h$$

6.2) given: u = eccentric anomaly
 f = true anomaly

$$r = a(1 - e \cos u)$$

$$n(t-T) = e \sin u - u \quad (8.2) \quad h > 0$$

$$r = a(1 - e \cos u)$$

(mean anomaly) $n(t-T) = u - e \sin u \quad (8.2) \quad h < 0$

$$r = \frac{a(1-e^2)}{1+e \cos f} \quad (8.5)$$

has (elliptical)

$$\text{trig identity: } \tan^2 \frac{u}{2} = \frac{1 - \cos u}{1 + \cos u}$$

$$\frac{a(1-e^2)}{1+e \cos f} = a(1 - e \cos u)$$

$$|e^2 - 1| = (1 - e \cos u)(1 + e \cos f)$$

$$|e^2 - 1| = 1 + e \cos f - e \cos u - e^2 \cos u \cos f$$

$$r = \frac{a(1-e^2)}{1+e \cos f} \quad \text{L.H.S. sign}$$

$$a \cos u = a e + r \cos f$$

$$a \cos u = a e + \frac{a(1-e^2) \cos f}{1+e \cos f}$$

$$\cos u = e + \frac{(1-e^2) \cos f}{1+e \cos f}$$

$$= e + \frac{(1-e^2) \cos f}{1+e \cos f}$$

$$\tan^2 \frac{u}{2} = \frac{1 - \cos u}{1 + \cos u} = \frac{1 - e - \frac{(1-e^2) \cos f}{1+e \cos f}}{1 + e + \frac{(1-e^2) \cos f}{1+e \cos f}}$$

$$= \frac{1 + e \cos f - e - \cos f}{1 + e \cos f + e + \cos f}$$

$$= \frac{1 + e \cos f - e - \cos f}{1 + e \cos f + e + \cos f}$$

$$\tan \frac{f}{2} = \frac{1 - \cos u}{1 + \cos u} = 1 \cdot \frac{1 - \cos f}{1 + \cos f} \cdot \frac{1 + \cos f}{1 - \cos f}$$

$$= \frac{1 - \cos f - e - \cos f}{1 + \cos f + e + \cos f} \cdot \frac{1 + \cos f}{1 - \cos f}$$

$$= \frac{1 + \cos f - e - \cos f}{1 + \cos f + e + \cos f}$$

$$\tan^2 f = \frac{1 - \cos f}{1 + \cos f}$$

$$\tan^2 u = \frac{(1 - \cos f) + e(\cos f - 1)}{(1 + \cos f) + e(\cos f + 1)}$$

$$= \frac{(1 - \cos f)(1 - e)}{(1 + \cos f)(1 + e)}$$

$$\tan^2 \frac{f}{2} = \frac{1 - e}{1 + e} \tan^2 \frac{f}{2}$$

$$\boxed{\tan \frac{f}{2} = \sqrt{\frac{1 - e}{1 + e}} \tan \frac{u}{2}, h < 0}$$

for $h > 0$ (hyperbolic)
use same process but for $\cosh / \tanh$

$$\Rightarrow \boxed{\tan \frac{f}{2} = \sqrt{\frac{1 + e}{1 - e}} \tanh \frac{u}{2}, h > 0}$$

8.4 show solutions are unique

$0 \leq e < 1$ elliptical
 $r(t - \tau) = u - e \sin u$
Mean anomaly
 $f(u) = u - e \sin u$
 $f'(u) = 1 - e \cos u \quad \cos u \in [-1, 1]$
 $e \cos u \leq e(1) \leq 1$
 $f'(u) = 1 - e \geq 0 \quad \forall u \Rightarrow$ function is monotonically increasing so only 1 solution for given t .

$e \geq 1$
 $f(u) = e \sinh u - u$
 $f'(u) = e \cosh u - 1 \quad \cosh(u) = 1 \quad \cosh u > 1 \quad \forall u \neq 0$
 $e \cosh u \geq e \cosh u \geq 1$
 $e \cosh u - 1 \geq 0 \quad \forall u \Rightarrow$ function is monotonically increasing

10.1 (10.2) $l = u - e \sin u \quad 0 \leq e < 1$
(10.1) $r = a(1 - e \cos u) \quad u - l = -e \sin u$

periodic: $u(l) = l$
(eccentric anomaly - mean anomaly)
 $u - l = e \sin u$
 $u - (l + 2\pi) = e \sin u$
 $\boxed{u - l = e \sin u + 2\pi = e \sin u}$

odd: $u(-l) = -u(l) \quad -l = -(u - e \sin u)$
 $-l = -(u - e \sin u)$
 $\boxed{u - l = -e \sin u}$

vanishes: $l = 0, l = \pi$
 $u - 0 = e \sin u$
 $\boxed{u - e \sin u \Rightarrow u = 0}$
 $u - \pi = e \sin u$
 $\boxed{u - e \sin u + \pi \Rightarrow u = \pi}$

cont. derivative
 $\frac{dl}{du} = \frac{d}{du}(u - e \sin u)$
 $1 - \frac{du}{du} - e \cos u \frac{du}{du}$
 $dl = (1 - e \cos u) du$
 $\frac{du}{dl} = \frac{1}{1 - e \cos u}$
 $\frac{d(u - l)}{dl} = \frac{du}{dl} - 1 = \frac{1}{1 - e \cos u} - 1$
 $= \frac{1 - (1 - e \cos u)}{1 - e \cos u} = \frac{e \cos u}{1 - e \cos u} \rightarrow \text{continuous}$
 $u(l) - l = \sum_{n=1}^{\infty} \frac{e^n}{n} u_n \sin n l$
continuous $\neq 0$

prove: $\frac{2}{\pi} \int_0^{\pi} \cos n(u - e \sin u) du$
fourier coefficients:
 $b_n = \frac{2}{\pi} \int_0^{\pi} f(u) \sin n u du$
 $u_n = \frac{2}{\pi} \int_0^{\pi} (u(l) - l) \sin n l dl$
integration by parts: $f = u(l) - l \quad g' = \sin n l$
 $f' = \frac{du}{dl} - 1 \quad g = -\frac{1}{n} \cos n l$
 $\frac{du}{dl} = \frac{1}{1 - e \cos u}$
 $f' = \frac{1}{1 - e \cos u} - 1$
 $f' = \frac{e \cos u}{1 - e \cos u}$
 $f' dl = \frac{e \cos u}{1 - e \cos u} (1 - e \cos u) du$
 $f' dl = \frac{e \cos u}{1 - e \cos u} du$
 $u_n = \frac{2}{\pi} \int_0^{\pi} \cos n l \cdot f' dl$
 $= \frac{2}{\pi} \int_0^{\pi} \cos n l \cdot \frac{e \cos u}{1 - e \cos u} du$
 $= \frac{2}{\pi} \int_0^{\pi} \cos(u - e \sin u) e \cos u du$
 $\boxed{u_n = \frac{2}{\pi} \int_0^{\pi} \cos(u - e \sin u) du}$

$$-l = (u - e \sin u)$$

10.2 given: q_0, q_1 pts on elliptical orbit
 u_0, u_1 eccentric anomalies
 $u_1 > u_0$
prove: distance $q_0 q_1$ is $2a \sin \alpha \sin \beta$

10.2 given: q_0, q_1 pts on elliptical orbit

u_0, u_1 eccentric anomalies
 $u_1 > u_0$

proj: distance $q_0 q_1$ is $2a \sin \alpha \sin \beta$

$$\alpha = \frac{1}{2}(u_1 - u_0) \quad \cos \beta = \cos \frac{1}{2}(u_1 + u_0)$$

$$\Delta t = \beta \cdot 2\pi \quad \gamma = \frac{1}{2}(u_1 + u_0)$$

$$\vec{r} = (a \cos u - ae, a \sqrt{1-e^2} \sin u)$$

$$|\vec{r}_1 - \vec{r}_2|^2 = a^2 (\cos u_1 - \cos u_0)^2 - a^2 (1-e^2) (\sin u_1 - \sin u_0)^2$$

$$(\cos u_1 - \cos u_0)^2 = \cos^2 u_1 - 2 \cos u_1 \cos u_0 + \cos^2 u_0$$

$$(\sin u_1 - \sin u_0)^2 = \sin^2 u_1 - 2 \sin u_1 \sin u_0 + \sin^2 u_0$$

$$|\vec{r}_1 - \vec{r}_2|^2 = \cos^2 u_1 + \cos^2 u_0 - 2 \cos u_1 \cos u_0 - (1-e^2) (\sin^2 u_1 + \sin^2 u_0 - 2 \sin u_1 \sin u_0)$$

$$\cos u_1 - \cos u_0 = -2 \sin \frac{u_1 + u_0}{2} \sin \frac{u_1 - u_0}{2}$$

$$\sin u_1 - \sin u_0 = 2 \cos \frac{u_1 + u_0}{2} \sin \frac{u_1 - u_0}{2}$$

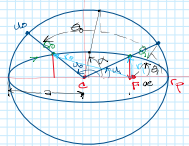
$$|\vec{r}_1 - \vec{r}_2|^2 = 4 \sin^2 \alpha (\sin^2 \gamma + (1-e^2) \cos^2 \gamma \sin^2 \gamma)$$

$$= 4 \sin^2 \alpha (\sin^2 \gamma + \cos^2 \gamma - \cos^2 \gamma \sin^2 \gamma)$$

$$= 4 \sin^2 \alpha (1 - \cos^2 \gamma \sin^2 \gamma)$$

$$|\vec{r}_1 - \vec{r}_2|^2 = 4 \sin^2 \alpha \sin^2 \beta \cos^2 \gamma$$

$$q_0 q_1 = 2 a \sin \alpha \sin \beta$$



$$10.2 \quad \vec{r} = -\frac{\mu}{r^2} \quad (10.2)$$

$$(\vec{r} \cdot \vec{r})' = r^2 = 2(\mu r + h r^2) \quad e > 0, \mu = G(M_1 + M_2)$$

$$(\vec{r} \cdot \vec{r})' = 2(\mu + h r) \quad r \rightarrow 0 \quad r' \rightarrow \frac{\mu}{h}$$

$$\text{Show } |\vec{r}(t) - \vec{r}(t_1)| \rightarrow \left(\frac{\mu}{2} \mu\right)^{1/2} \text{ as } t \rightarrow t_1$$

$$r^2 = 2\mu + 2h r^2$$

$$\sqrt{r^2} = \sqrt{2\mu} \rightarrow \sqrt{2\mu}$$

$$|\vec{r}|^{1/2} \rightarrow \sqrt{2\mu} \quad \checkmark$$

$$\frac{dr}{dt} = -\sqrt{\frac{2\mu}{r}}$$

$$\int \frac{dr}{\sqrt{2\mu}} = \int \frac{dt}{\sqrt{2\mu}}$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu} t + C$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu} (t - t_1)$$

$$r^{3/2} = \frac{3}{2} \sqrt{2\mu} (t - t_1)$$

$$r = \left[\frac{3}{2} \sqrt{2\mu} (t - t_1) \right]^{2/3}$$

$$r = \left(\frac{3}{2} \mu \right)^{1/3} (t - t_1)^{2/3}$$

$$\frac{r}{(t - t_1)^{2/3}} = \left(\frac{3}{2} \mu \right)^{1/3}$$

$$r(t - t_1)^{-2/3} \rightarrow \left(\frac{3}{2} \mu \right)^{1/3} \text{ as } t \rightarrow t_1$$

$$10.1 \quad \vec{r} = \frac{\mu}{r} \vec{e} + \frac{F_a \vec{a}}{r}, \quad \frac{F_a \vec{a}}{r} = r \times \left(-\frac{\mu}{r^3} \vec{e} \right) + \frac{F_a \vec{a}}{r}$$

$$\vec{e} = r \frac{F_a \vec{a}}{r} = \frac{F_a \vec{a}}{r} \quad 10.1 \quad \mu \vec{e} = \vec{r} \times \vec{e} + \frac{F_a \vec{a}}{r} \cdot \left[(\mu \vec{e} \cdot \vec{r}) + \frac{F_a \vec{a}}{r} \right] \vec{r}$$

$$\mu = \frac{F_a \vec{a}}{r} \sin \phi \quad \vec{r} \times \vec{e} = -\alpha \quad \vec{e} \times \vec{e} = 0$$

$$\vec{e} = \vec{r} \times \left(\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \right) \quad \vec{r} \times \vec{e} = -\alpha \quad \vec{e} \times \vec{e} = 0$$

$$= -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{r} \quad \alpha = r \cdot e$$

$$\vec{e} = -\frac{F_a \vec{a}}{r} + \frac{r F_a \vec{e}}{r}$$

$$\mu \vec{e} = \left(\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \right) \times \vec{e} = -\frac{F_a \vec{a}}{r} \times \vec{e} + \frac{F_a \vec{a}}{r} \times \vec{e} + \frac{F_a \vec{a}}{r} \times \vec{e}$$

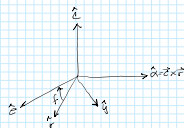
$$= -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e}$$

$$= -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e}$$

$$\mu \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e}$$

$$\mu \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e}$$

$$\mu \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e} = -\frac{F_a \vec{a}}{r} + \frac{F_a \vec{a}}{r} \cdot \vec{e}$$



Hwk 1 Prob 2

Friday, January 23, 2026 10:20 PM

2) prove identities

Linear momentum:

$$\begin{aligned}
 \vec{F}_{ij} &= \frac{Gm_i m_j}{r_{ij}^3} \vec{r}_{ji} = -\frac{Gm_i m_j}{r_{ji}^3} \vec{r}_{ij} \\
 \sum_{i=1}^N \vec{F}_i &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{Gm_i m_j}{r_{ij}^3} \vec{r}_{ji} \\
 &= \sum \left(\frac{m_i m_j}{r_{ij}^3} \vec{r}_{ji} + \frac{m_i m_j}{r_{ji}^3} \vec{r}_{ij} \right) \\
 &= \sum \frac{m_i m_j}{r_{ij}^3} (\vec{r}_{ji} + \vec{r}_{ij}) = 0 \\
 \text{so } \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_{ji} &= 0
 \end{aligned}$$

angular momentum

$$\vec{X} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \times \vec{r}_{ji} = -\sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_j \times \vec{r}_{ij}$$

$$\begin{aligned}
 2\vec{X} &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \left(\frac{m_i m_j}{r_{ij}^3} \vec{r}_i \times \vec{r}_{ji} + \frac{m_i m_j}{r_{ij}^3} \vec{r}_j \times \vec{r}_{ij} \right) \\
 &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} (\vec{r}_i \times \vec{r}_{ji} + \vec{r}_j \times \vec{r}_{ij}) \\
 &= \sum_{i=1}^N \sum_{j=1}^N \frac{m_i m_j}{r_{ij}^3} (\vec{r}_i \times (\vec{r}_i - \vec{r}_j) + \vec{r}_j \times (\vec{r}_j - \vec{r}_i)) \\
 &= \sum_{i=1}^N \sum_{j=1}^N \frac{m_i m_j}{r_{ij}^3} (-\vec{r}_{ij} - \vec{r}_{ji}) = 0
 \end{aligned}$$

$$\begin{aligned}
 2\vec{X} &= 0 \\
 \vec{X} &= 0
 \end{aligned}$$

$$\text{so } \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \times \vec{r}_{ji} = 0$$

Energy

$$\sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \frac{m_i m_j}{r_{ij}} = 2 \sum_{j=1}^N \sum_{i=1}^N \frac{m_i m_j}{r_{ij}}$$

$$U = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \frac{m_i m_j}{r_{ij}} = -G \left[2 \sum_{i < j \leq N} \frac{m_i m_j}{r_{ij}} \right]$$

$$U = -G \sum_{i < j \leq N} \frac{m_i m_j}{r_{ij}}$$

total energy

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$\frac{dT}{dt} = \frac{1}{2} \sum_{i=1}^N m_i (\dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i + \ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i)$$

$$\frac{dT}{dt} = \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i$$

$$\vec{F}_i = m_i \ddot{\vec{r}}_i = \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ji}$$

$$\frac{dT}{dt} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\frac{dU}{dt} = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{d}{dt} \left(\frac{m_i m_j}{\|\vec{r}_{ij}\|} \right)$$

$$= -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j \left(\frac{1}{\|\vec{r}_{ij}\|^2} \right) \frac{d \cdot \|\vec{r}_{ij}\|}{dt} \rightarrow \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}$$

$$= +G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j \frac{1}{\|\vec{r}_{ij}\|^2} \left(\frac{1}{2 \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}} \right) \frac{d(\vec{r}_{ij} \cdot \vec{r}_{ij})}{dt}$$

$$= -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} (\dot{\vec{r}}_i - \dot{\vec{r}}_j) \cdot \vec{r}_{ij}$$

$$\frac{dU}{dt} = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ij}$$

$$\frac{d(T+U)}{dt} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\frac{d}{dt} - \sum_{l=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{G m_i m_j}{r_{ij}^3} \vec{r}_i \cdot \vec{r}_{ji} = 0$$

$$\boxed{\frac{d(T+U)}{dt} = 0}$$

$$3) U = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|}$$

$$U = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\alpha \vec{r}_{ij}\|} = -\frac{G}{\alpha} \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|} = \boxed{-G \alpha^{-1} \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|} \Rightarrow n = -1}$$

$$\frac{\partial U}{\partial \vec{r}_i} = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \cdot \vec{r}_{ij}$$

$$\vec{r}_i \cdot \frac{\partial U}{\partial \vec{r}_i} = G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}^3} \underbrace{\vec{r}_i \cdot \vec{r}_i}_{r_i^2} \cdot \vec{r}_{ij}$$

$$\boxed{\vec{r}_i \cdot \frac{\partial U}{\partial \vec{r}_i} = G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} \vec{r}_{ij} = -U}$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i \alpha \dot{\vec{r}}_i \cdot \alpha \dot{\vec{r}}_i = \boxed{\alpha^2 \left(\frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i \right) \Rightarrow n = 2}$$

$$\frac{\partial T}{\partial \vec{r}_i} = \sum_{i=1}^N m_i \dot{\vec{r}}_i$$

$$\boxed{\vec{r}_i \cdot \frac{\partial T}{\partial \vec{r}_i} = \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i = 2T}$$

$$I_p = \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i$$

$$I_p = \sum_{i=1}^N m_i \alpha \vec{r}_i \cdot \alpha \vec{r}_i = \boxed{\alpha^2 r_i \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i \Rightarrow N = 2}$$

$$\frac{\partial I_p}{\partial \vec{r}_i} = 2 \sum_{i=1}^N m_i \vec{r}_i$$

$$\boxed{\vec{r}_i \cdot \frac{\partial I_p}{\partial \vec{r}_i} = 2 \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i = 2I_p}$$

$$I_p U^2 = \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i \left[G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} \right]^2$$

$$\boxed{= \sum_{i=1}^N m_i \alpha \vec{r}_i \cdot \alpha \vec{r}_i \left(G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\cancel{\alpha r_{ij}}} \right)^2 \Rightarrow n = 0}$$

for homogeneous functions order product of functions equals sum of order $(2 + 2(-1) = 0)$
so order = 0

$$4) \vec{H} = \sum_{i=1}^N m_i \vec{r}_i \times \dot{\vec{r}}_i$$

$$H^2 \leq 2T I_P$$

$$a = \vec{r} \quad b = \dot{\vec{r}}$$

$$\vec{H} = \sum m_i a \times b$$

$$\vec{H}^2 = \sum m_i a \times b \sum m_i a \times b$$

$$= \sum m_i^2 (a \times b) (a \times b)$$

$$H^2 = \sum m_i^2 [a^2 b^2 - (a \cdot b)^2]$$

$$H^2 \leq 2T I_P$$

$$T = \frac{1}{2} \sum m_i a^2$$

$$I_P = \sum m_i b^2$$

$$A = 2T$$

$$B = I_P$$

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5)

Lagrange Identity

$$\left(\sum_{k=1}^N a_k^2\right) \left(\sum_{k=1}^N b_k^2\right) - \left(\sum_{k=1}^N a_k b_k\right)^2 = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (a_i b_j - a_j b_i)^2$$

$$I_P = \sum_{i=1}^N m_i r_i^2 = \sum_{i=1}^N m_i (\vec{r}_i \cdot \vec{r}_i)$$

$$\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (\vec{r}_i - \vec{r}_j)^2$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (r_i^2 - 2\vec{r}_i \cdot \vec{r}_j + r_j^2)$$

$$\frac{1}{2M} \left(\sum_{i=1}^N m_i r_i^2 \sum_{j=1}^N m_j - 2 \sum_{i=1}^N \sum_{j=1}^N \vec{r}_i \cdot \vec{r}_j + \sum_{j=1}^N m_j \sum_{i=1}^N m_i r_j^2 \right)$$

$$\underbrace{\frac{1}{2} \sum_{i=1}^N m_i r_i^2}_{I_P} - \underbrace{\frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i \cdot \sum_{j=1}^N m_j \vec{r}_j}_{M \vec{R}_c \cdot \vec{R}_c} + \underbrace{\frac{1}{2M} \sum_{i=1}^N m_i \sum_{j=1}^N m_j r_j^2}_{I_P}$$

$$\frac{1}{2} I_P + \frac{1}{2} I_P - M \vec{R}_c \cdot \vec{R}_c$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j \vec{r}_i \cdot \vec{r}_j = I_P - M \vec{R}_c \cdot \vec{R}_c$$

$$I_P = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j \vec{r}_i \cdot \vec{r}_j + M \vec{R}_c \cdot \vec{R}_c$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i v_i^2$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_{ij}^2 = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (v_i - v_j)^2$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (v_i^2 - 2\vec{v}_i \cdot \vec{v}_j + v_j^2)$$

$$= \frac{1}{2M} \left(\sum_{i=1}^N \sum_{j=1}^N m_i m_j v_i^2 - \sum_{i=1}^N \sum_{j=1}^N 2m_i m_j \vec{v}_i \cdot \vec{v}_j + \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_j^2 \right)$$

$$= \frac{1}{2M} \left(\underbrace{\sum_{i=1}^N m_i v_i^2 \sum_{j=1}^N m_j}_{I_P} - \sum_{i=1}^N m_i \vec{v}_i \cdot \sum_{j=1}^N m_j \vec{v}_j + \underbrace{\sum_{j=1}^N m_j v_j^2 \sum_{i=1}^N m_i}_{I_P} \right)$$

$$= \frac{1}{2M} \left(\underbrace{\sum_{i=1}^N m_i v_i^2}_T \underbrace{\sum_{j=1}^N m_j}_M \underbrace{\sum_{i=1}^N m_i \vec{v}_i \cdot \sum_{j=1}^N m_j \vec{v}_j}_{\vec{v}_c} + \underbrace{\sum_{i=1}^N m_i}_M \underbrace{\sum_{j=1}^N m_j v_j^2}_T \right)$$

$$= \frac{1}{2} (2T - M \vec{v}_c \cdot \vec{v}_c)$$

$$T = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_{ij}^2 + \frac{1}{2} M \vec{v}_c \cdot \vec{v}_c$$

6) show:

$$\sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N f_{ij} = \frac{1}{2} \sum_{1 \leq i < j \leq N} f_{ij} ; f_{ii} = 0$$

$$\sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i = \frac{1}{M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N m_i m_j \vec{r}_{ij} \cdot \vec{v}_{ij}$$

$$\vec{r}_i \cdot \vec{v}_i \text{ at } i=j \quad \vec{r}_{ij} \cdot \vec{v}_{ij} = 0$$

$$\frac{1}{M} \left[\frac{1}{2} \sum_{1 \leq i < j \leq N} m_i m_j (\vec{r}_j - \vec{r}_i) \cdot (\vec{v}_j - \vec{v}_i) \right]$$

$$\begin{aligned} & \frac{1}{2M} \sum_{1 \leq i < j \leq N} m_i m_j [\vec{r}_j \cdot \vec{v}_j - \vec{r}_j \cdot \vec{v}_i - \vec{r}_i \cdot \vec{v}_j + \vec{r}_i \cdot \vec{v}_i] \\ &= \frac{1}{2M} \left[\left(\sum_{j=1}^N m_j \vec{r}_j \cdot \vec{v}_j \right) \left(\sum_{i=1}^N m_i \right) - \underbrace{\left(\sum_{j=1}^N m_j \vec{r}_j \right)}_{M \vec{R}_c} \cdot \underbrace{\left(\sum_{i=1}^N m_i \vec{v}_i \right)}_{M \vec{V}} - \underbrace{\left(\sum_{i=1}^N m_i \vec{r}_i \right)}_{M \vec{R}_c} \cdot \underbrace{\left(\sum_{j=1}^N m_j \vec{v}_j \right)}_{M \vec{V}} \right] \end{aligned}$$

$$\sum_{j=1}^N m_j \vec{r}_j \cdot \vec{v}_j = \sum m_i \vec{r}_i \cdot \vec{v}_i$$

$$= \frac{1}{2M} (2M \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i - 2M \vec{R}_c \cdot M \vec{V})$$

$$= \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i - M \vec{R}_c \cdot \vec{V} \quad \checkmark \quad \text{at COM } \vec{R}_c = 0$$

$$\boxed{\sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i = \frac{1}{M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N m_i m_j \vec{r}_{ij} \cdot \vec{v}_{ij}}$$

show:

$$\sum_{i=1}^N m_i [r_i^2 \vec{U} - \vec{r}_i \vec{r}_i] = \frac{1}{M} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j [r_{ij}^2 \vec{U} - \vec{r}_{ij} \vec{r}_{ij}]$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N m_i m_j [r_{ij}^2 \vec{U} - \vec{r}_{ij} \vec{r}_{ij}]$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N m_i m_j [(r_j^2 - 2\vec{r}_i \cdot \vec{r}_j + r_i^2) \vec{U} - (\vec{r}_j \vec{r}_j - \vec{r}_j \vec{r}_i - \vec{r}_i \vec{r}_j + \vec{r}_i \vec{r}_i)]$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N \underbrace{m_i m_j}_{\frac{m_i m_j}{2\vec{r}_i \vec{U} - \vec{r}_i \vec{r}_i}} \left[\underbrace{r_i^2 \vec{U} - \vec{r}_i \vec{r}_i}_{\frac{r_i^2 \vec{U} - \vec{r}_i \vec{r}_i}{2\vec{r}_i \vec{U} - \vec{r}_i \vec{r}_i}} + \underbrace{[r_j^2 \vec{U} - \vec{r}_j \vec{r}_j]}_{\frac{[r_j^2 \vec{U} - \vec{r}_j \vec{r}_j]}{2\vec{r}_i \vec{U} - \vec{r}_i \vec{r}_i}} - 2(\vec{r}_i \cdot \vec{r}_j) \vec{U} + \vec{r}_i \vec{r}_j + \vec{r}_j \vec{r}_i \right]$$

$$= \sum_{i=1}^N r_i^2 \vec{U} - \vec{r}_i \vec{r}_i + \frac{1}{M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N m_i m_j [-2(\vec{r}_i \cdot \vec{r}_j) \vec{U} + \vec{r}_i \vec{r}_j + \vec{r}_j \vec{r}_i]$$

$$= \sum_{i=1}^N r_i^2 \vec{U} - \vec{r}_i \vec{r}_i - \frac{1}{M} \left[\left(\sum_{i=1}^N m_i \vec{r}_i \right) \left(\sum_{j=1, j \neq i}^N m_j \vec{r}_j \right) \vec{U} + \sum_{i=1}^N m_i \vec{r}_i \sum_{j=1, j \neq i}^N m_j \vec{r}_j + \sum_{j=1}^N m_j \vec{r}_j \sum_{i=1, i \neq j}^N m_i \vec{r}_i \right]$$

$$= \sum_{i=1}^N r_i^2 \vec{U} - \vec{r}_i \vec{r}_i - \frac{1}{M} \underbrace{M \vec{R}_c M \vec{R}_c \vec{U}}_0 + 2 \underbrace{M \vec{R}_c \vec{R}_c}_0$$

$$\text{at COM } \vec{R}_c = 0$$

$$\boxed{\sum_{i=1}^N r_i^2 \vec{U} - \vec{r}_i \vec{r}_i = \frac{1}{M} \sum_{i=1}^N \sum_{j=1, j \neq i}^N m_i m_j [r_{ij}^2 \vec{U} - \vec{r}_{ij} \vec{r}_{ij}]}$$

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$$7) M_1 = m_1 \quad M_2 = m_1 + m_2 \quad M_3 = m_1 + m_2 + m_3$$

$$\vec{R}_1 = \vec{r}_2 - \vec{r}_1 \quad \vec{r} = \vec{R}_2 = \vec{r}_3 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \quad R_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{M}$$

$$T = \frac{1}{2} \sum_{i=1}^3 m_i \dot{\vec{r}}_i^2$$

define $M_i = \sum_{j=1}^N m_j$; $\vec{R}_N \equiv$ system center of mass

$$T = \frac{1}{2} \sum_{i=1}^{N-1} M_{i-1} \frac{M_{i-1} m_i}{M_i} \dot{\vec{R}}_i \cdot \dot{\vec{R}}_i + \frac{1}{2} M_N \dot{\vec{R}}_N \cdot \dot{\vec{R}}_N$$

$$T = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \dot{\vec{R}} \cdot \dot{\vec{R}} + \frac{1}{2} \frac{m_3 (m_1 + m_2)}{m_1 + m_2 + m_3} \dot{\vec{r}} \cdot \dot{\vec{r}} \quad \text{KE of COM}$$

$$\mu_1 = \frac{m_1 m_2}{M_2} \quad \mu_2 = \frac{M_2 m_3}{M_3}$$

$$\text{COM: } \vec{R}_1 = \vec{r}_2 - \vec{R}_1^c = \vec{r}_2 - \vec{r}_1$$

$$\vec{R}_2 = \vec{r}_3 - \vec{R}_2^c = \vec{r}_3 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$\vec{r}_3 = \vec{R}_2^c + \vec{r}$$

$$\vec{r}_2 = \vec{R}_2^c + \frac{m_1}{M_2} \vec{R}_1$$

$$\vec{r}_1 = \vec{R}_2^c - \frac{m_2}{M_2} \vec{R}_1$$

$$U = - \frac{G m_1 m_2}{|\vec{R}|} - \frac{G m_1 m_3}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|} - \frac{G m_2 m_3}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|}$$

$$L = T - U$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{R}}} = - \frac{\partial U}{\partial \vec{R}}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{R}}} = \frac{m_1 m_2}{M_2} \ddot{\vec{R}}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{r}}} = \frac{m_3 M_2}{M_3} \ddot{\vec{r}}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_1 m_2}{R} \right) = - G m_1 m_2 \frac{\vec{R}}{R^3}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_1 m_3}{\vec{r} + \frac{m_2}{M_2} \vec{R}} \right) = - G m_1 m_3 \frac{m_2}{M_2} \frac{(\vec{r} + \frac{m_2}{M_2} \vec{R})}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|^3}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_2 m_3}{\vec{r} - \frac{m_1}{M_2} \vec{R}} \right) = - G m_2 m_3 \frac{(-m_1)}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3} = \frac{G m_1 m_2 m_3}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3}$$

$$- \frac{\partial U}{\partial \vec{R}} = \left[- \frac{G m_1 m_2}{R^3} \vec{R} - \frac{G m_1 m_2 m_3}{M_2} \frac{(\vec{r} + \frac{m_2}{M_2} \vec{R})}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|^3} - \frac{G m_2 m_3}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3} \right] = \frac{m_1 m_2}{M_2} \ddot{\vec{R}}$$

$$-\frac{\partial U}{\partial \vec{R}} = \left[-\frac{Gm_1m_2}{R^3} \vec{R} - \frac{Gm_1m_2m_3}{M_2} \left[\frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{\left| \vec{S} + \frac{m_3}{M_2} \vec{R} \right|^3} - \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{\left| \vec{S} - \frac{m_1}{M_2} \vec{R} \right|^3} \right] \right] = \frac{m_1m_2}{M_2} \vec{R}$$

$$-\frac{\partial U}{\partial \vec{S}} \left(-\frac{Gm_1m_3}{\left| \vec{S} + \frac{m_3}{M_2} \vec{R} \right|^3} \right) = -Gm_1m_3 \frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{\left| \vec{S} + \frac{m_3}{M_2} \vec{R} \right|^3}$$

$$-\frac{\partial U}{\partial \vec{S}} \left(-\frac{Gm_2m_3}{\left| \vec{S} - \frac{m_1}{M_2} \vec{R} \right|^3} \right) = -Gm_2m_3 \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{\left| \vec{S} - \frac{m_1}{M_2} \vec{R} \right|^3}$$

$$\frac{\partial U}{\partial \vec{S}} = \frac{m_3M_2}{M_3} \ddot{\vec{S}} = -Gm_1m_3 \frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{\left| \vec{S} + \frac{m_3}{M_2} \vec{R} \right|^3} - Gm_2m_3 \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{\left| \vec{S} - \frac{m_1}{M_2} \vec{R} \right|^3}$$

$N=4$ Jacobi

$$M_4 = m_1 + m_2 + m_3 + m_4$$

$$\vec{R}_3 = \vec{r}_4 + \vec{R}_3^c = \vec{r}_4 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{M_3}$$

$N=5$ Jacobi

$$M_5 = M_4 + m_5$$

$$\vec{R}_4 = \vec{r}_5 + \vec{R}_4^c = \vec{r}_5 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3 + m_4\vec{r}_4}{M_4}$$